

#### 2.4. Possible choices of $\Delta$

The class of possible differences in trends  $\Delta$  must be specified by the researcher, and the choice of  $\Delta$  will depend on the economic context. We highlight several possible choices of  $\Delta$  that may be reasonable in empirical applications and formalize intuitive arguments that are commonly made by applied researchers regarding possible violations of parallel trends. Throughout our discussion, we write  $\delta_{pre} = (\delta_{-T}, \dots, \delta_{-1})'$  and  $\delta_{post} = (\delta_1, \dots, \delta_{\bar{T}})'$ , with  $\delta_0$  normalized to zero. This aligns the notation with Example 1, where  $\delta$  corresponds to the difference in trends between treated and comparison groups, and  $\delta_0$  is normalized to zero.

**2.4.1. Bounding relative magnitudes.** In empirical applications, researchers may be willing to assume that the confounding factors which produce non-parallel trends in the post-treatment periods are not too much larger in magnitude than the confounding factors in the pre-treatment periods. In their empirical application to right-to-carry gun laws, [Manski and Pepper \(2018\)](#) operationalize this intuition by calibrating bounds on  $|\delta_1|$  to the largest violations of parallel trends in the pre-treatment period (see their Table 3).<sup>6</sup> Such a restriction can be formalized in our framework by imposing that  $\delta \in \Delta^{RM}(\bar{M})$  for  $\bar{M} \geq 0$ , where

$$\Delta^{RM}(\bar{M}) = \{\delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M} \cdot \max_{s < 0} |\delta_{s+1} - \delta_s|\}.$$

$\Delta^{RM}(\bar{M})$  bounds the maximum post-treatment violation of parallel trends between consecutive periods by  $\bar{M}$  times the maximum pre-treatment violation of parallel trends. We use the abbreviation RM for “relative magnitudes.” The choice  $\Delta^{RM}(\bar{M})$  may be reasonable if the researcher suspects that possible violations of parallel trends are driven by confounding economic shocks that are of a similar magnitude to confounding economics shocks in the pre-period. When the number of pre-treatment and post-treatment periods is similar, a natural benchmark may be  $\bar{M} = 1$ , which bounds the worst-case post-treatment difference in trends by the equivalent maximum in the pre-treatment period.<sup>7</sup>

**2.4.2. Example: non-staggered DiD (continued).** In the three-period DiD model ( $T = \bar{T} = 1$ ), assuming  $\delta \in \Delta^{RM}(\bar{M}) = \{(\delta_{-1}, \delta_1)' : |\delta_1| \leq \bar{M}|\delta_{-1}|\}$  bounds the magnitude of  $\delta_1$  based on the magnitude of  $\delta_{-1}$ . The larger the magnitude of the pre-treatment violation in parallel trends,  $|\delta_{-1}|$ , the wider the range of possible post-treatment violations of parallel trends. ▲

**2.4.3. Smoothness restrictions.** In other empirical settings, researchers may be worried about confounding from secular trends (*e.g.* long-run changes in labour supply) that they suspect evolve smoothly over time. In such settings, it is common for empirical researchers to control for a linear group-specific time trend.<sup>8</sup> This approach is valid if the difference in trends is linear,

6. In their application, [Manski and Pepper \(2018\)](#) observe the outcome for the entire population of interest, and thus their observed pre-treatment data corresponds with  $\delta_{pre}$  rather than  $\hat{\beta}_{pre}$ .

7. In settings where later pre-periods are thought to be more informative, the researcher could use a different value of  $\bar{M}$  for different pre-periods, *e.g.* imposing that  $|\delta_{t+1} - \delta_t| \leq \max\{|\delta_0 - \delta_{-1}|, 2 \cdot |\delta_{-1} - \delta_{-2}|\}$ . Likewise, although we’ve considered bounds on changes across consecutive periods, the researcher could also impose bounds on changes across multiple periods, *e.g.*  $|\delta_{t+2} - \delta_t| \leq \bar{M} \cdot \max_{s < -1} |\delta_{s+2} - \delta_s|$ .

8. Specifically, researchers often augment specification (2) with group-specific linear trends, an approach [Dobkin et al. \(2018\)](#) refer to as a “parametric event-study.” An analogous approach is to estimate a linear trend using only observations prior to treatment, and then subtract out the estimated linear trend from the observations after treatment ([Bhuller et al., 2013](#); [Goodman-Bacon, 2018, 2021](#)).

i.e.  $\Delta = \{\delta : \delta_t = \gamma \cdot t, \gamma \in \mathbb{R}\}$ . There are often concerns, however, that the linear specification is not exactly correct (Wolfers, 2006; Lee and Solon, 2011). A natural relaxation is therefore to impose only that the differential trends evolve smoothly over time by bounding the extent to which its slope may change across consecutive periods. Such a restriction can be formalized in our framework by imposing that  $\delta \in \Delta^{\text{SD}}(M)$  for  $M \geq 0$ , where

$$\Delta^{\text{SD}}(M) := \{\delta : |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq M, \forall t\}. \quad (8)$$

The parameter  $M \geq 0$  governs the amount by which the slope of  $\delta$  can change between consecutive periods, and thus bounds the discrete analogue of the second derivative. We use the abbreviation SD for “second differences” or “second derivative.”<sup>9</sup> In the special case where  $M = 0$ ,  $\Delta^{\text{SD}}(0)$  requires that the difference in trends be exactly linear, which corresponds with the assumption underlying the parametric linear specification common in applied work.

**2.4.4. Example: non-staggered DiD (continued).** In the three-period DiD model, assuming the differential trend is exactly linear is equivalent to assuming  $\Delta = \{\delta : \delta_1 = -\delta_{-1}\}$ . Assuming  $\delta \in \Delta^{\text{SD}}(M)$  requires only that the linear extrapolation be *approximately* correct,  $\delta_1 \in [-\delta_{-1} - M, -\delta_{-1} + M]$ .

**2.4.5. Combining smoothness and relative magnitudes bounds.** In some contexts, researchers may be willing to assume that the difference in trends evolves relatively smoothly over time but may be unsure about the smoothness bound  $M \geq 0$  introduced above. In such cases, it may be reasonable to assume that the possible non-linearities in the post-treatment difference in trends are bounded by the observed non-linearities in the pre-treatment difference in trends. This can be formalized with the restriction

$$\Delta^{\text{SDRM}}(\bar{M}) = \{\delta : \forall t \geq 0, |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq \bar{M} \cdot \max_{s < 0} |(\delta_{s+1} - \delta_s) - (\delta_s - \delta_{s-1})|\},$$

which bounds the maximum deviation from a linear trend in the post-treatment period by  $\bar{M} \geq 0$  times the equivalent maximum in the pre-treatment period. The set  $\Delta^{\text{SDRM}}(\bar{M})$  is thus similar to  $\Delta^{\text{SD}}(M)$  introduced above, except it allows the magnitude of the possible non-linearity to explicitly depend on the observed pre-trends.

**2.4.6. Sign and monotonicity restrictions.** Context-specific knowledge may sometimes also suggest sign or monotonicity restrictions on the differential trend. For instance, if the policy of interest occurs at the same time as a confounding policy change that we expect to have a positive effect on the outcome, we might restrict the post-treatment bias to be positive,  $\delta \in \Delta^{\text{PB}} := \{\delta : \delta_t \geq 0 \forall t \geq 0\}$ . Likewise, there may be secular pre-existing trends that we expect would have continued following the treatment date.<sup>10</sup> We may then wish to impose that the differential trend be increasing,  $\delta \in \Delta^{\text{I}} := \{\delta : \delta_t \geq \delta_{t-1} \forall t\}$ , or monotone with unknown sign,  $\delta \in \Delta^{\text{Mon}} := \Delta^{\text{I}} \cup (-\Delta^{\text{I}})$ . Sign and monotonicity restrictions may be combined with the

9. Restrictions on the second derivative of the conditional expectation function or density have been used in regression discontinuity settings (Kolesár and Rothe, 2018; Frandsen, 2016; Noack and Rothe, 2020). Smoothness restrictions are also used to obtain partial identification in e.g. Manski (1997); Kim *et al.* (2018).

10. Monotone violations of parallel trends are often discussed in applied work. For example, Lovenheim and Willen (2019a) argue that violations of parallel trends cannot explain their results because “pre-[treatment] trends are either zero or in the wrong direction (i.e. opposite to the direction of the treatment effect).” Greenstone and Hanna (2014) estimate upward-sloping pre-existing trends and argue that “if the pre-trends had continued” their estimates would be upward biased.

previously discussed restrictions, such as  $\Delta^{SDPB}(M) := \Delta^{SD}(M) \cap \Delta^{PB}$ ,  $\Delta^{SDI}(M) := \Delta^{SD}(M) \cap \Delta^I$ , and  $\Delta^{RMI}(\bar{M}) := \Delta^{RM}(\bar{M}) \cap \Delta^I$ .

**2.4.7. Polyhedral restrictions.** Although the restrictions described above will be sensible in many empirical contexts, researchers will often have context-specific knowledge that motivates alternative restrictions than what we introduced above. To accommodate such cases, we consider the broad class of  $\Delta$ s that can be written as polyhedra (sets defined by linear inequalities), or the finite union of polyhedra.

**Definition 1** (Polyhedral restriction). The class  $\Delta$  is polyhedral if it takes the form  $\Delta = \{\delta : A\delta \leq d\}$  for some known matrix  $A$  and vector  $d$ .

All of the examples described above can be written either as polyhedral restrictions or finite unions of such restrictions. For instance,  $\Delta^{SD}(M)$  and  $\Delta^{SDPB}(M)$  can be written directly as polyhedra.<sup>11</sup> Likewise,  $\Delta^{RM}(\bar{M})$  and  $\Delta^{SDRM}(\bar{M})$  can be written as the finite union of polyhedra, where each polyhedron corresponds with a different location for the maximum pre-treatment violation.<sup>12</sup>

The class of (finite unions of) polyhedra is quite broad, and allows for a variety of other restrictions that may be relevant in empirical work. For example, researchers studying labour market training and related programs may be concerned about Ashenfelter's dip (Ashenfelter, 1978), in which earnings for the treated group trend downwards (relative to control) before treatment and upwards afterwards. In this type of setting, researchers might naturally use a polyhedral  $\Delta$  to impose (i) restrictions on the signs of the pre-treatment and post-treatment biases, as well as (ii) restrictions on the magnitude of the rebound effect relative to the pre-treatment shock.

## 2.5. Inferential goal

As discussed above, the event study coefficients  $\hat{\beta}_n$  will satisfy  $\sqrt{n}(\hat{\beta}_n - \beta) \rightarrow_d \mathcal{N}(0, \Sigma^*)$  for a wide variety of commonly used estimators. This suggests the finite-sample normal approximation

$$\hat{\beta}_n \approx_d \mathcal{N}(\beta, \Sigma_n), \quad (9)$$

where  $\approx_d$  denotes approximate equality in distribution and  $\Sigma_n = \Sigma^*/n$ . We will construct confidence sets that are uniformly valid for all parameter values  $\theta$  in the identified set when the approximation in (9) holds exactly with  $\Sigma_n$  known. That is, we construct confidence sets  $\mathcal{C}_n(\hat{\beta}_n, \Sigma_n)$  satisfying

$$\inf_{\delta \in \Delta, \tau} \inf_{\theta \in \mathcal{S}(\delta + \tau, \Delta)} \mathbb{P}_{\hat{\beta}_n \sim \mathcal{N}(\delta + \tau, \Sigma_n)} \left( \theta \in \mathcal{C}_n(\hat{\beta}_n, \Sigma_n) \right) \geq 1 - \alpha. \quad (10)$$

In Section 3.3, we will show that finite-sample size control in the normal model in the sense of (10) translates to uniform asymptotic size control over a large class of data-generating processes when  $\Sigma_n$  is replaced by a consistent estimate  $\hat{\Sigma}_n$ . That is, we will show that the constructed confidence sets further satisfy

11. In our ongoing three-period DiD example,  $\Delta^{SD}(M) = \{\delta : A^{SD}\delta \leq d^{SD}\}$  for  $A^{SD} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$  and  $d^{SD} = (M, M)'$ . This generalizes naturally when there are multiple pre-periods and multiple post-periods.

12. For example, define the polyhedra  $\Delta_{s,+}^{RM}(\bar{M}) = \{\delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M}(\delta_{s+1} - \delta_s)\}$  and  $\Delta_{s,-}^{RM} = \{\delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq -\bar{M}(\delta_{s+1} - \delta_s)\}$ . Then  $\Delta^{RM}(\bar{M}) = \bigcup_{s < 0} (\Delta_{s,+}^{RM}(\bar{M}) \cup \Delta_{s,-}^{RM}(\bar{M}))$ .

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}} \inf_{\theta \in S(\delta_P + \tau_P, \Delta)} \mathbb{P}_P \left( \theta \in \mathcal{C}_n(\hat{\beta}_n, \hat{\Sigma}_n) \right) \geq 1 - \alpha. \quad (11)$$

for a large class of distributions  $\mathcal{P}$  such that  $\delta_P \in \Delta$  for all  $P \in \mathcal{P}$ .

We will focus on constructing confidence sets for the case where  $\Delta$  is a polyhedron. For the case where  $\Delta$  is the finite union of polyhedra, a valid confidence set can be constructed by taking the union of the confidence sets for each of its components.

**Lemma 2.2.** *Suppose that for each  $k = 1, \dots, K$ , the confidence set  $\mathcal{C}_{n,k}(\hat{\beta}_n, \Sigma_n)$  satisfies (10) with  $\Delta = \Delta_k$ . Then the confidence set  $\mathcal{C}_n(\hat{\beta}_n, \Sigma_n) = \bigcup_{k=1}^K \mathcal{C}_{n,k}(\hat{\beta}_n, \Sigma_n)$  satisfies (10) with  $\Delta = \bigcup_{k=1}^K \Delta_k$ .*

In the next two sections, we introduce two approaches to obtain confidence sets satisfying (10), with different desirable properties depending on the form of  $\Delta$ . The first approach, based on moment inequalities, accommodates a wide range of restrictions  $\Delta$  and has some desirable asymptotic power guarantees. The second approach, based on FLCIs, can potentially offer finite-sample power improvements for certain special classes of  $\Delta$  of interest, such as  $\Delta^{\text{SD}}(M)$ .

### 3. INFERENCE USING MOMENT INEQUALITIES

In this section, we introduce a general approach for inference that has good asymptotic properties over a large class of possible restrictions  $\Delta$ . We show that inference on the partially identified parameter  $\theta = l' \tau_{\text{post}}$  in this setting is equivalent to testing a system of moment inequalities with a potentially large number of nuisance parameters that enter the moments linearly. We consider an implementation based on the conditional approach developed in ARP, which allows us to obtain computationally tractable confidence sets with desirable power properties for many parameter configurations.

#### 3.1. Representation as a moment inequality problem with linear nuisance parameters

Consider the problem of conducting inference on  $\theta = l' \tau_{\text{post}}$  when  $\Delta$  takes the polyhedral form  $\Delta = \{\delta : A\delta \leq d\}$ . We will develop tests that control size under the null hypothesis  $H_0 : \theta = \bar{\theta}, \delta \in \Delta$  when the normal approximation (9) holds exactly with known variance matrix  $\Sigma_n$ . In Section 3.3, we will provide conditions under which size control in the finite sample normal model translates to uniform asymptotic size control over a large class of data-generating processes.

As a first step, we show that testing  $H_0$  is equivalent to testing a system of moment inequalities with linear nuisance parameters in the normal model. Observe that if  $\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)$  for  $\beta$  satisfying (3), then  $\mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)}[\hat{\beta}_n - \tau] = \delta$ . It follows that  $\delta \in \Delta = \{\delta : A\delta \leq d\}$  if and only if  $\mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)}[A\hat{\beta}_n - A\tau] \leq d$ . Defining  $Y_n = A\hat{\beta}_n - d$  and  $L_{\text{post}} = [0, I']$  to be the matrix such that  $\tau = L_{\text{post}} \tau_{\text{post}}$ , it is immediate that the null hypothesis  $H_0$  is equivalent to the composite null

$$H_0 : \exists \tau_{\text{post}} \in \mathbb{R}^{\tilde{T}} \text{ s.t. } l' \tau_{\text{post}} = \bar{\theta} \quad \text{and} \quad \mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)} [Y_n - AL_{\text{post}} \tau_{\text{post}}] \leq 0. \quad (12)$$

Testing the null hypothesis  $H_0$  is therefore equivalent to testing that the moment inequalities  $\mathbb{E}_{\hat{\beta}_n \sim \mathcal{N}(\beta, \Sigma_n)} [Y_n - AL_{\text{post}} \tau_{\text{post}}] \leq 0$  hold for some value of  $\tau_{\text{post}}$  satisfying  $l' \tau_{\text{post}} = \bar{\theta}$ .